



12. INTRODUCTION TO CONDUCTIVITY PROBE

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Conductivity tester is a simple, but very important instrument, which is able to test for faults many components like: diodes, transistors, coils, transformers, speakers and headphones, capacitors, switches, jumpers, cables and many other different electronic components. This method is a lot faster and straightforward than it is using some “off the shelf” instrument.

Schematic for this device is on 12.1a. It is called a relaxation audio oscillator. When you connect points A and B using a piece of copper wire, a variable current flows through the transistors as sequences of impulses. This means that immediately upon connecting the points A and B, current level rapidly rises to some destined maximum value, and then drops to zero. For certain amount of time there is no current, after which it rises again rapidly, and whole cycle repeats itself. Since relation of times when current is flowing and when it is not is highly in favor of the later, this kind of current is called the spike impulse current. Collector current of a T2 transistor flows through the speaker which generates sound, whose base frequency could be calculated using this approximate equation.

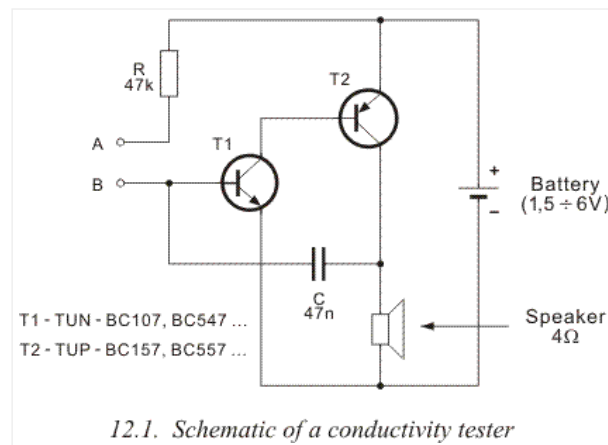
$$f = \frac{1,6}{RC}$$

In our case $R=47 \text{ k}\Omega$ and $C=47 \text{ nF}$, which means:

$$f = \frac{1,6}{47 \cdot 10^3 \cdot 47 \cdot 10^{-9}} = 724 \text{ Hz.}$$

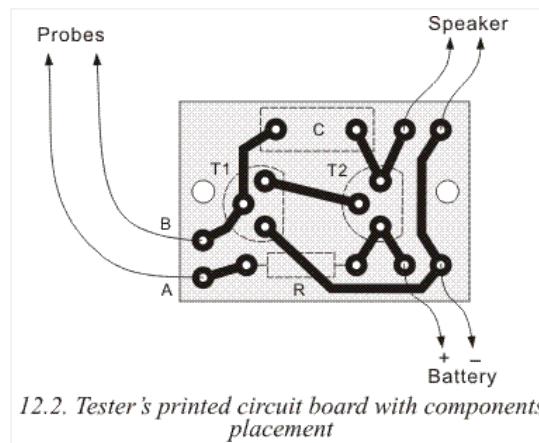
From the equation above, it is clear that varying of the frequency is possible by varying the resistor or capacitor value. Frequency rise is achieved by lowering the resistance or capacitance of the circuit, and vice versa, rising the values of the resistor or capacitor, lowers the oscillator frequency. Active variation of the frequency base is possible by replacing the resistor with a several hundred kilohm trimmer potentiometer. If such modification of the circuit was needed, special care must be taken not to set the trimmer into it's lowest position since this means zero resistance, and that could burn the transistors. To avoid unnecessary care and further complicating the operation of this straightforward device, low value resistor could be connected to the trimmer in series. This resistor would act as a protection for transistors inside of the circuit since it facilitates a minimum resistance, and thus doesn't leave transistors bare in the frying pan when the trimmer is in it's lowest position.

In this example we used an 1.5V battery for supply, but it is possible to plug this instrument on any battery between 1.5V and 9V.

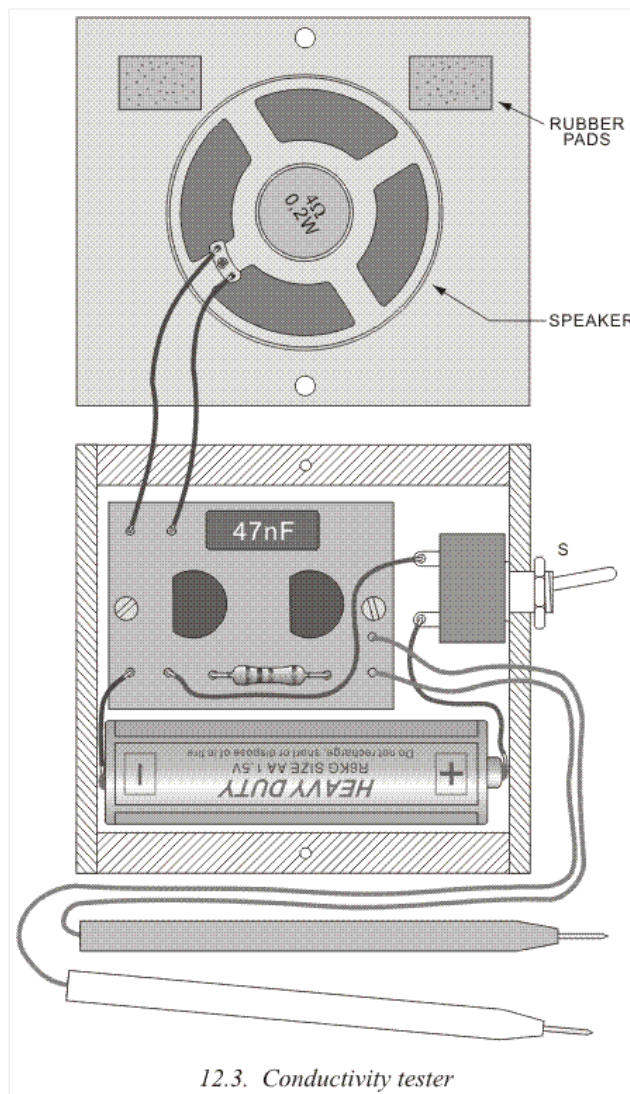


Current flowing through the component that is being tested is lower than $I=U/R$, where U is the voltage of the supply battery, and R is the resistance of the resistor in the base circuit. In our example, these values are $U=1,5\text{ V}$ and $R=47\text{ k}\Omega$, which means that current flow is $I=32\text{ micro amperes}$, which is very low, so tested component is safe from harm from this device.

Oscillator's printed board design is on 12.2. This is viewed from the copper plated side of the board, components are placed on the other side, so their positions are marked in dotted lines. Component side of the board is on 12.2.



Printed board, battery and the speaker are placed in a small box, as shown on 12.3. Miniature speaker is fixed to the upper pane of the box using two wood screws. It is connected to the circuit board using two threaded isolated wires. Same wires are used for all other connections as well. Battery is connected to the board using these wires, for example. In our example, wires are soldered directly to the poles of the battery, and the board fixed inside of the box using wood screws and two rectangular wooden pads glued to the bottom of the box, leaving just enough space to squeeze the battery in. These are not proper solutions, they are cheap "hacks" used when other options are limited. But these are functional for people who always have their trusty soldering iron at hand. What would be a proper solution? Buying a battery holder (with enough battery slots as needed) or battery clips (for those square 9V batteries) would simplify the process of changing the battery, although this circuit is very low in power consumption. Other thing is plastic or metal mounts for boards, these are pretty cheap and you should keep them at hand in your "junk box" when experimenting with electronics.



12.3. Conductivity tester

On the front side of the box, we drilled two holes, one for the switch and the other for wires which hold probes on their ends. Probes are cheap components and come in various shapes and sizes with various purposes in mind. Since we've been applying dirty methods, like soldering the battery, there is no reason why we should back from building our own probes now. Any old marker-pen will do, just slip thicker copper wire through it's center, and sand/grind/cut protruding ends into a pointy tip. It is advisable to make probes in different colors, red and black are dominant standards for distinguishing them. Positive probe (red) is connected to point A, and negative one (black) is connected to the point B. You could use alligator clips instead of probes, for example, this would leave your hands free for other purposes, but for some precise testing of the on-board components, go with the more precise probes we already mentioned.

Give your new instrument the initial self-test (battery might be empty, or some other unexpected thing happened) by connecting the probe tips together. If sound is heard from the speaker, everything is fine and ready for work.

Ok, everything is working, now you want to play with your new toy. Check, for example, conductivity of your own body. Hold probe tips between thumb and index finger of your left and right hand. What you hear is a sound whose level and especially frequency depend on your skin moistness. Wow, now instrument could be used as a very crude and inaccurate lie detector. This probably wouldn't be accepted in a court of law, but may be an interesting game you play on your friends. "Suspect" holds in his/hers hands probes which could be made of a metal pipe for this occasion. Pipe should be wide enough so that a large portion of palm surface is actually in contact with metal. When the suspect starts dodging questions or lying, his palms start sweating more than usual, and the tone produced by our "lie detector" is higher than usual.



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